

Variables, Data Types, User Input, Operators & Simple Programs

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1. Introduction to Programming Basics

Programming is the process of giving instructions to a computer to perform specific tasks. To write programs effectively, we must understand basic concepts such as:

- Variables
- Data Types
- User Input
- Operators
- Arithmetic Logic

These concepts form the foundation of all programming languages.

2. Variables

What is a Variable?

A **variable** is a named storage location used to store data in a program. The value stored inside a variable can change during program execution.

Variables help programmers store and manipulate information.

Example

```
name = "Rahul"
```

```
age = 20
```

```
marks = 85
```

Here:

- name stores text data
 - age stores numerical data
 - marks stores numerical data
-

Why Variables are Important

Variables are used to:

- Store user information
 - Perform calculations
 - Save temporary results
 - Manage program data efficiently
-

Real-Life Example

Think of a variable as a labeled container.

Example:

- Water Bottle → Stores water
 - School Bag → Stores books
 - Variable → Stores data
-

3. Rules for Naming Variables

Variable names should follow certain rules:

Rules

- Must begin with a letter or underscore _
 - Cannot start with a number
 - Cannot contain spaces
 - Cannot use special symbols except _
 - Variable names are case-sensitive
-

Valid Variable Names

student_name
age
_marks
totalAmount

Invalid Variable Names

1name
student name
total-amount

4. Data Types

What are Data Types?

A **data type** defines the type of data a variable can store.

Different types of data require different storage methods and operations.

Common Data Types

4.1 Integer (int)

Stores whole numbers.

Example

age = 18

4.2 Float (float)

Stores decimal numbers.

Example

```
price = 99.99
```

4.3 String (str)

Stores text or characters.

Example

```
name = "InternStudio"
```

4.4 Boolean (bool)

Stores only two values:

- True
- False

Example

```
isPassed = True
```

Data Type Table

Data Type Description		Example
int	Whole numbers	10
float	Decimal values	12.5
str	Text data	"Hello"
bool	True/False values	True

5. Type Conversion

What is Type Conversion?

Type conversion means changing one data type into another.

Example

```
age = "20"  
newAge = int(age)
```

Here, string "20" is converted into integer 20.

Common Conversion Functions

Function Purpose

`int()` Convert to integer

`float()` Convert to float

`str()` Convert to string

`bool()` Convert to boolean

6. User Input

What is User Input?

User input allows users to enter data while the program is running.

Example

```
name = input("Enter your name: ")  
print(name)
```

Input with Integer Conversion

```
age = int(input("Enter age: "))  
print(age)
```

Importance of User Input

User input makes programs:

- Interactive
 - Dynamic
 - User-friendly
-

7. Operators

What are Operators?

Operators are symbols used to perform operations on variables and values.

Types of Operators

1. Arithmetic Operators
 2. Comparison Operators
 3. Logical Operators
 4. Assignment Operators
-

8. Arithmetic Operations

Arithmetic operators perform mathematical calculations.

Arithmetic Operators Table

Operator	Meaning	Example
+	Addition	$A + B$
-	Subtraction	$A - B$
*	Multiplication	$A * B$
/	Division	A / B
%	Modulus	$A \% B$
//	Floor Division	$A // B$
**	Exponent	$A ** B$

Example Program

```
a = 10
b = 5

print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

9. Logical Operations

Logical operators are used to combine conditions.

Logical Operators Table

Operator Meaning

and	True if both conditions are true
or	True if at least one condition is true
not	Reverses the condition

Example

```
age = 20
```

```
print(age > 18 and age < 30)
```

10. Comparison Operators

Comparison operators compare two values.

Comparison Operators Table

Operator Meaning

==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

Example

```
a = 10
```

```
b = 20
```

```
print(a < b)
```

11. Assignment Operators

Assignment operators assign values to variables.

Assignment Operators Table

Operator	Example
=	x = 5
+=	x += 2
-=	x -= 2
*=	x *= 2
/=	x /= 2

Example

```
x = 10
```

```
x += 5
```

```
print(x)
```

12. Simple Programs

Program 1: Addition of Two Numbers

```
a = int(input("Enter first number: "))
```

```
b = int(input("Enter second number: "))
```

```
sum = a + b
```

```
print("Sum =", sum)
```

Program 2: Area of Rectangle

```
length = float(input("Enter length: "))
```

```
breadth = float(input("Enter breadth: "))
```

```
area = length * breadth
```



```
print("Area =", area)
```

Program 3: Even or Odd

```
num = int(input("Enter number: "))
```

```
if num % 2 == 0:
```

```
    print("Even")
```

```
else:
```

```
    print("Odd")
```

Program 4: Simple Calculator

```
a = int(input("Enter first number: "))
```

```
b = int(input("Enter second number: "))
```

```
print("Addition =", a + b)
```

```
print("Subtraction =", a - b)
```

```
print("Multiplication =", a * b)
```

```
print("Division =", a / b)
```

13. Real-Life Applications

These concepts are used in:

- Banking software
- Calculator applications
- Login systems
- Mobile apps
- Websites
- Artificial Intelligence
- Data analysis systems